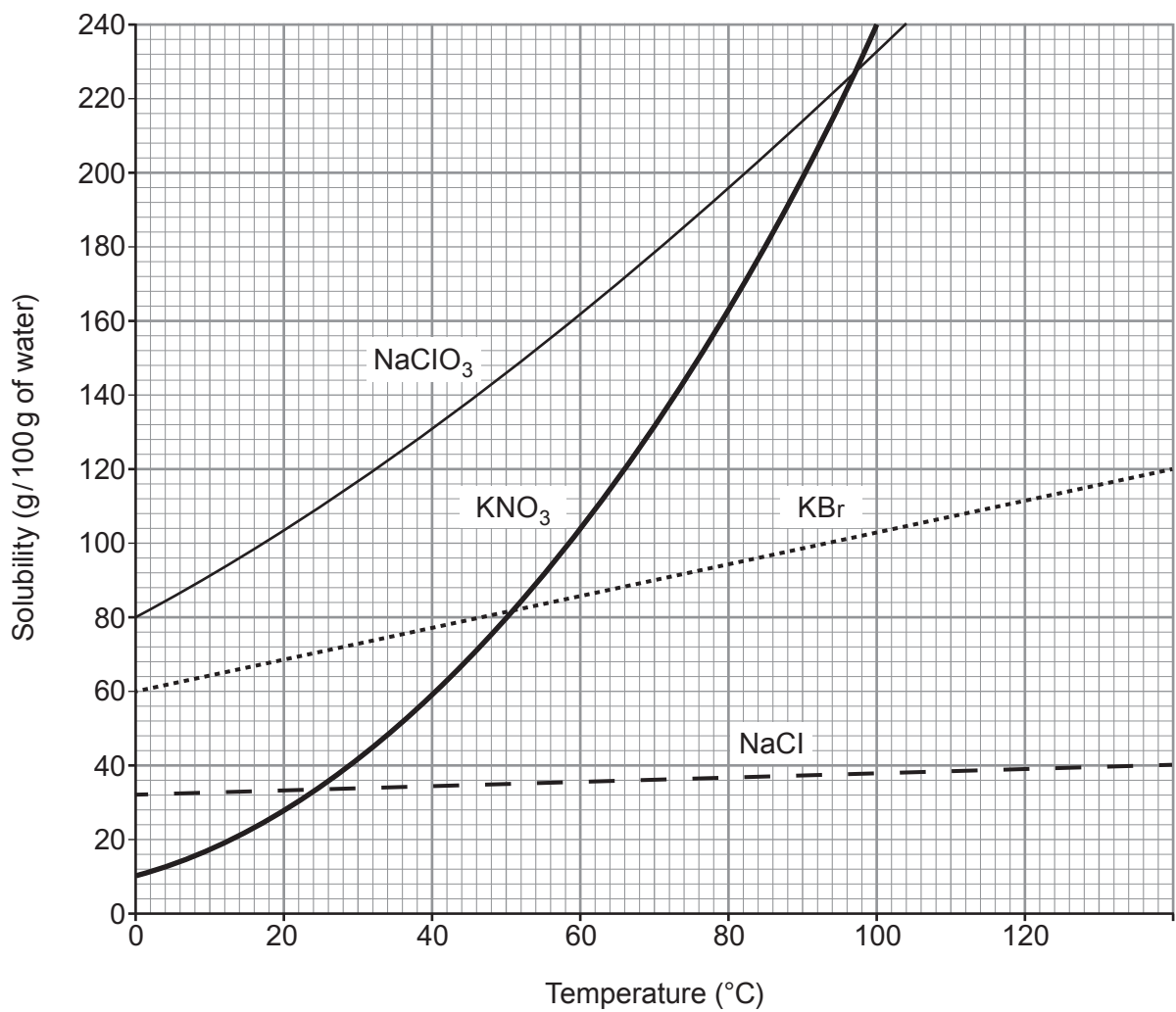


10. The grid below shows the solubility curves for four ionic compounds.



- NaClO₃ sodium chlorate
- KNO₃ potassium nitrate
- KBr potassium bromide
- NaCl sodium chloride



- (a) (i) Give the temperature at which the solubility of potassium nitrate and potassium bromide is the same. [1]

..... °C

- (ii) Calculate the mass of solid potassium nitrate that would form if a saturated solution in 200 g of water were cooled from 100 °C to 20 °C. [3]

Mass = g

- (iii) Suggest why a student may be surprised at the temperature range shown on the solubility curves. [1]

.....
.....

- (b) (i) Give the symbols of the **ions** of Group 1 elements present in the compounds shown on the grid. [1]

.....

- (ii) Explain how these ions are formed from their atoms. [2]

.....
.....

- (c) Potassium nitrate reacts with aluminium hydroxide to produce aluminium nitrate and potassium hydroxide.

Balance the symbol equation for the reaction taking place. [1]

